Chromium Browser Artifact Parser

Investigator's Guide to Reviewing Output

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1. Introduction

Purpose of This Guide

This guide is designed to assist forensic investigators, incident responders, and security analysts in effectively reviewing and interpreting the Excel output generated by the Chromium Browser Artifact Parser. Understanding how to properly analyze this data is crucial for drawing accurate conclusions in investigations.

What the Parser Extracts

The Chromium Browser Artifact Parser extracts forensic artifacts from Chromium-based browsers (Google Chrome, Microsoft Edge, Opera, Brave, Vivaldi, etc.) and presents them in a structured Excel workbook. Each worksheet contains a specific category of browser data, complete with decoded timestamps and descriptive field values.

Key Investigation Principles

- Timestamps are in UTC — All decoded timestamps are in Coordinated Universal Time. Convert to local time as needed.
- Absence of evidence is not evidence of absence — Missing data could indicate deletion, data that was never created, or data that aged out of retention.
- Correlate across worksheets — A single artifact often tells only part of the story. Cross-reference timestamps and URLs across multiple worksheets to build a complete picture.
- Consider data retention limits — Chromium browsers automatically purge certain data. Understanding these limits is critical for proper interpretation.

2. Understanding the Excel Output

Workbook Structure

The Excel output typically contains the following worksheets (in order):

- Summary — Overview of record counts for each artifact category, plus tool version
- Preferences — User identity and browser configuration data
- Artifact Worksheets — Individual sheets for each processed artifact type

Note: The History Clusters artifact produces six worksheets rather than one.

The Summary Worksheet

The Summary worksheet should be your first stop. It provides a quick overview of record counts, and a record count of 0 may indicate the user cleared that data type or the artifact doesn't exist. Unusually low counts (compared to expected usage) may suggest data deletion. The script version is also recorded at the bottom of the Summary worksheet.

Column Naming Convention

- Raw timestamp columns — Named after the database field (e.g., visit_time, date_created)
- Decoded timestamp columns — Prefixed with 'Decoded' (e.g., Decoded visit_time (UTC))
- Decoded value columns — Prefixed with 'Decoded' for enumerated values (e.g., Decoded danger_type)

3. Critical Data Retention Information

The 90-Day History Limitation

Chromium-based browsers automatically delete browsing history older than approximately 90 days. This means the History worksheet will only contain visits from roughly the last 3 months. You cannot conclude that a user never visited a site if it does not appear in history — they may have visited it more than 90 days ago, and that history has been automatically purged by the browser, not necessarily by user action.

Data That Can Persist Beyond 90 Days

Worksheet	Retention Period	Notes
History	~90 days	Automatically purged
History Clusters	~90 days	Derived from History database; subject to same retention
Downloads	Indefinite	Until user clears or database limit reached
Login Data	Indefinite	Saved credentials persist until deleted
Autofill	Indefinite	Form data persists until deleted
Bookmarks	Indefinite	Persist until manually deleted
Cookies	Varies	Based on expiration or session
Addresses	Indefinite	Until manually deleted
Credit Cards	Indefinite	Until manually deleted
Bank Accounts	Indefinite	Until manually deleted
Loyalty Cards	Indefinite	Until manually deleted
Extensions	Indefinite	Present as long as extension is installed
WebAssist (Edge)	Potentially years	Separate database, independent retention policy

Special Case: Microsoft Edge WebAssist

The WebAssist worksheet (Edge only) can contain browsing data far older than 90 days — potentially spanning years of activity. This is because WebAssist is a separate database from the main History database used by Edge's AI assistant features, and it maintains its own navigation history independent of the standard history retention policy. If you are analyzing a Microsoft Edge profile, the WebAssist worksheet may be your only source of browsing activity from more than 90 days ago.

4. Worksheet Reference Guide

4.1 History Worksheet

Source Database: History. Contains complete browsing history with navigation details.

Column	Description	Investigative Value
urls.id	Unique URL identifier	Links to other tables
visits.id	Unique visit identifier	NULL if no corresponding visit
url	The visited URL	Primary evidence of site access
title	Page title	Context for what user viewed
visit_count	Total visits to this URL	Frequency of access
typed_count	Times URL was manually typed	Indicates intentional navigation
Visit Source	Local, Synced, Extension, Imported	Origin of the record
Transition Type	How user navigated to page	User action context
Decoded visit_time (UTC)	When visit occurred	Timeline evidence
visit_duration	Time spent on page (seconds)	User engagement level

4.2 Downloads Worksheet

Source Database: History. Contains downloaded file information including target path, referrer, tab URL, timestamps, download source URL, danger type, and completion state.

4.3 Search Terms Worksheet

Source Database: History + Web Data. Contains search queries performed by the user. Note: this worksheet primarily captures searches through recognized search engines (Google, Bing, etc.). Searches performed through on-site search boxes may not appear here — check the History worksheet for URLs containing query parameters such as ?q= or ?search=.

4.4 Bookmarks Worksheet

Source Database: Bookmarks (JSON) + Bookmarks.bak (JSON). Combines data from both the current Bookmarks file and its backup. The backup may contain bookmarks that were subsequently deleted, revealing recently removed items or historical interests the user may have tried to conceal.

4.5 Login Data Worksheet

Source Database: Login Data. Contains saved website credentials including domain, username, creation date, last used date, and usage count. Password values are encrypted and cannot be decrypted by this tool.

4.6 Cookies Worksheet

Source Database: Network/Cookies. Contains browser cookies with creation, last access, and expiration timestamps.

4.7 Autofill Worksheet

Source Database: Web Data. Contains form field auto-complete data including field name, user-entered value, usage count, and first/last used dates. Common fields include email, username, name, address, phone, and company.

4.8 Addresses Worksheet

Source Database: Web Data. Contains saved address book entries including full name, email, phone, street address, city, state, postal code, and country. Particularly valuable for identifying the user and their known locations.

4.9 Credit Cards Worksheet

Source Database: Web Data. Contains masked credit card information (last four digits only), including card network, expiration date, and usage stats.

4.10 Bank Accounts Worksheet

Source Database: Web Data. Contains masked bank account information with usage metadata. Nicknames may reveal personal information or account purposes. Usage data shows when and how often accounts were used for autofill.

4.11 Loyalty Cards Worksheet

Source Database: Web Data. Contains store loyalty and rewards card information. Loyalty card numbers can be used (with proper legal process) to obtain purchase history from retailers. Regional retailers can establish geographic patterns.

4.12 Shortcuts Worksheet

Source Database: Shortcuts. Contains omnibox (address bar) shortcuts including the typed text, resulting URL, hit count, and last access time.

4.13 Top Sites Worksheet

Source Database: Top Sites. Shows sites that appear on Chrome's new tab page — the user's most frequently visited destinations.

4.14 FavIcons Worksheet

Source Database: Favicons. Contains website icons and associated URLs. The page_url column can reveal sites visited even if history was cleared, since favicon data can persist after history deletion.

4.15 Keywords Worksheet

Source Database: Web Data. Shows configured search engines. The Decoded date_created (UTC) indicates when a search engine was first used, revealing when a user first performed a search on a particular site.

4.16 Extensions Worksheet

New in version 2026-Mar-1. Source: Extensions/ subfolder manifest.json files. Contains the installed extension inventory with ID, Name, Version, Description, Author, and Homepage URL.

Column	Description	Investigative Value
ID	Unique extension identifier	Cross-reference with Chrome Web Store
Name	Extension display name	Identify tool or capability

Column	Description	Investigative Value
Version	Installed version	Identify potentially outdated or unusual versions
Description	Extension purpose	Understand function (e.g., VPN, ad blocker, download manager)
Author	Developer or publisher	Identify source and legitimacy
Homepage URL	Extension website	Research extension background

Investigative Significance: Extensions can download files, intercept traffic, modify web content, or exfiltrate data. Identifying installed extensions can reveal capabilities the user had available, explain unusual network activity, or corroborate other findings.

4.17 WebAssist Worksheet (Edge Only)

Source Database: WebAssistDatabase. Contains Edge-specific navigation history. As discussed in Section 3, this worksheet can contain data far older than the 90-day history limit. For Edge investigations, this may be your primary source of historical browsing evidence.

5. Understanding Gap Analysis Worksheets

What Are Gap Worksheets?

The parser includes three Gap worksheets: History Gaps, Downloads Gaps, and Login Data Gaps. These worksheets identify missing records in the database by detecting gaps in the sequential record numbering (ROWID). SQLite databases assign incrementing IDs to records, so gaps in the sequence indicate that records once existed but have been deleted.

How to Read Gap Data

Column	Description
Previous Record Number	The ID of the record that exists BEFORE the gap
Record Number	The ID of the record that exists AFTER the gap
Number of Missing Records	How many records are missing in this gap
Beginning Timestamp (UTC)	Timestamp of the record BEFORE the gap
Ending Timestamp (UTC)	Timestamp of the record AFTER the gap

What Gaps Can Tell You

- Evidence of Deletion — Gaps indicate records were deleted, either by the user manually clearing browsing data, by third-party cleaning software, or through targeted deletion of specific entries
- Timeframe of Deleted Activity — The beginning and ending timestamps bracket when the deleted activity occurred
- Volume of Deleted Data — The number of missing records indicates how much data was deleted in that gap
- Pattern Analysis — Multiple gaps with similar timestamps may indicate periodic clearing

Limitations of Gap Analysis

- Gap analysis cannot tell you what was deleted, only when and how many
- If a user clears all browsing data, the database is reset and gaps from that clearing will not be detected
- Some gaps may result from browser crashes or corruption, not intentional deletion
- The Downloads database does not track its sequence number in sqlite_sequence, so gaps at the end cannot be detected for downloads

6. Understanding History Clusters Worksheets

What Are History Clusters?

Chrome's History database includes a set of cluster tables introduced alongside the Journeys feature. Chrome's content analysis pipeline groups related browsing visits into 'clusters' representing research sessions or topical browsing activity. Even though the Journeys UI was removed from Chrome 125 (May 2024), the underlying cluster tables continue to be populated in modern Chrome builds.

When the History Clusters artifact is selected, the parser produces six separate worksheets rather than one. Each worksheet examines the cluster data from a different forensic angle.

Important Note on cluster_keywords

The cluster_keywords table was designed to store per-cluster keyword labels extracted by Chrome's keyword pipeline. This table is empty in current Chrome builds because the keyword extraction pipeline was disabled before the Journeys UI was removed. The Clusters Search Term worksheet therefore derives search term data from the content_annotations table (which is actively populated) rather than cluster_keywords.

The Six Cluster Worksheets

Clusters Overview

Lists all clusters with their labels, keywords (where available from cluster_keywords), prominent flag, and total visit count. Use as an initial triage to understand the scope and topics of clustered browsing activity.

Clusters Contents

Shows every visit within each cluster with decoded timestamps, URLs, page titles, cluster relevance score, engagement score, URL variants stored by the clustering engine, visit duration, and full transition details. This is the core worksheet for reconstructing what the user viewed during each research session.

Clusters Search Term

Identifies search terms associated with cluster visits, derived from the content_annotations table. For each cluster and search term combination, shows the number of visits, the temporal span (earliest and latest visit timestamps plus span in minutes), and a list of associated URLs. HIGH forensic value for establishing user intent.

Clusters Timeline

Provides a temporal overview of each cluster: first and last visit timestamps (raw and decoded), session duration in minutes, total visit count, and unique URL count. Useful for establishing when a research session occurred and how long it lasted.

Clusters Duplicate Visits

Shows pairs of visits that Chrome's clustering engine identified as duplicates — visits to the same logical page made during a session. Repeated visits to the same page indicate sustained user interest and may indicate the page was particularly relevant to the user's research or intent. Columns include both the canonical and duplicate visit timestamps, URLs, titles, and the time gap between visits.

Clusters Comprehensive Export

A full join of cluster data with content_annotations (search terms, page categories, entities, related searches, visibility score) and context_annotations (foreground duration, duration since last visit, page end reason). This is the most complete single-worksheet export, suitable for ingestion into forensic timeline tools.

Investigative Use of Cluster Data

- Establish user intent — The Clusters Search Term worksheet directly shows what topics the user was researching
- Reconstruct sessions — The Clusters Contents worksheet shows the exact sequence of pages viewed within a research session
- Identify sustained interest — Duplicate visit data shows pages the user returned to repeatedly
- Timeline correlation — Cross-reference cluster timestamps with History, Downloads, and Login Data to build a complete activity picture
- Content annotations — Page categories and entities (where populated) can reveal the topical nature of browsing even without reading the actual pages

7. The Extensions Worksheet

Overview

The Extensions worksheet is produced by scanning the Extensions/ subfolder within the browser profile. For each installed extension, the parser reads the manifest.json file from the latest version subfolder and extracts key metadata.

Columns

Column	Description	Investigative Value
ID	Unique Chrome extension identifier (32-character string)	Look up in Chrome Web Store; flag unofficial or sideloaded extensions
Name	Display name from manifest	Identify the extension's function
Version	Installed version number	Identify outdated versions or unknown builds
Description	Self-reported purpose from manifest	Understand capability (download manager, VPN, script injector, etc.)
Author	Developer field from manifest	Assess legitimacy and source
Homepage URL	Developer website	Research the extension's background

Investigative Significance

- Data exfiltration tools — Extensions with broad permissions (access to all sites, file downloads, clipboard) can exfiltrate data
- Anonymization — VPN or proxy extensions may explain gaps in other evidence or unusual traffic patterns
- Download managers — May explain downloaded files not captured in the standard Downloads table
- Sideloaded extensions — Extensions not from the Chrome Web Store may indicate unauthorized software or malware
- Correlation with history — If an extension's homepage URL appears in the History worksheet, it may indicate the user was actively managing or researching that extension

8. The Preferences Worksheet: User Identity Evidence

Overview

The Preferences worksheet extracts data from the browser's Preferences JSON file. This is one of the most valuable worksheets for identifying the user associated with a browser profile.

Key Evidence Types

User Identity Information

If the user was logged into their Google account (Chrome) or Microsoft account (Edge), you will find email, full name, given name, GAIA ID (a unique Google Account identifier that persists across sessions), and profile picture URL.

Profile Information

Includes profile name, creation date, browser version at profile creation, and previous navigation time (the last navigation before the browser was closed).

User Behavior Settings

Includes default download directory, last Save As directory, startup behavior, URL startup list, homepage, and new tab shortcuts. These can reveal the user's preferred destinations and recent file activity.

Exit Status

A 'Crashed' exit type may indicate the browser was closed unexpectedly, the system crashed, the browser process was terminated by task manager, or potentially suspicious termination to hide activity.

9. Cross-Referencing Data Across Worksheets

Technique: Activity Context from History

When you find an interesting artifact in any worksheet with a timestamp, navigate to the History worksheet and filter for activity around that same time to reveal what the user was browsing when the artifact was created.

Technique: Correlating Cluster Sessions with Other Artifacts

Use the Clusters Timeline worksheet to identify the start and end times of a browsing session. Then filter the History, Downloads, Login Data, and Cookies worksheets for that same timeframe to build a complete picture of what the user did during that session.

Technique: Identifying Related Accounts

Cross-reference Login Data (saved credentials), Autofill (email addresses and usernames), Addresses (personal information), Preferences (logged-in account), and Cookies (sites with persistent sessions) to identify all accounts associated with the user.

Technique: Verifying Claimed Activity

Check History for the visit, Cookies for session evidence, FavIcons for site access evidence, Shortcuts for address bar activity, and Gap Analysis to see if relevant timeframes show deletion.

Technique: Building User Profiles from Searches

Combine Search Terms (direct search queries), Clusters Search Term (search terms within sessions), Keywords (search engines used), History (URLs with query parameters), and Autofill (search terms in form fields).

Technique: Assessing Extension Risk

Review the Extensions worksheet, then check History for visits to the extensions' homepage URLs, check Downloads for files that may have been facilitated by download-manager extensions, and check Shortcuts for extension-related navigation patterns.

10. Interpreting Missing Worksheets

Why a Worksheet Might Be Missing

- Zero records exist — The database table is empty
- Database file doesn't exist — The expected file isn't in the profile
- Processing error — An error occurred (check the Summary for '0' counts)
- Not selected for processing — The artifact wasn't checked in the parser

Investigative Interpretations

Missing Worksheet	Possible Explanations
History	User cleared all history; new profile; history disabled
History Clusters	No cluster data yet formed; browser version does not populate cluster tables
Downloads	User never downloaded files; user cleared downloads
Login Data	User doesn't save passwords; user cleared saved passwords
Cookies	User cleared cookies; cookies disabled; privacy mode used
Search Terms	No recognized searches performed; search history cleared
Extensions	No extensions installed; Extensions folder not found
WebAssist	Not an Edge profile; Edge feature not used

Annex A: Investigative Tips and Techniques

A.1 Essential Excel Skills for Browser Forensics

- Filtering: Use AutoFilter (Ctrl+Shift+L) on each worksheet. Use 'Contains' filters for partial URL matching.
- Sorting: Sort by timestamp columns for chronological views. Sort by visit_count or number_of_hits to find frequently accessed content.
- Conditional Formatting: Highlight rows with specific keywords or color-code timestamps within investigation timeframes.
- Find and Replace (Ctrl+H): Quickly locate specific domains or terms using wildcards for pattern matching.

A.2 URL Filtering Strategies

To Find	Filter For	Column to Filter
Gmail inbox access	mail.google.com/mail containing inbox	url
Google Drive activity	drive.google.com	url
YouTube viewing	youtube.com/watch	url
Facebook	facebook.com	url
Twitter/X	twitter.com or x.com	url
LinkedIn	linkedin.com	url
Outlook web	outlook.live.com or outlook.office.com	url
OneDrive	onedrive.live.com	url
Executables downloaded	.exe	target_path
Archives downloaded	.zip, .rar, .7z	target_path

A.3 Timeline Analysis Techniques

- Creating a Master Timeline: Copy timestamp and key columns from multiple worksheets into a new sheet, add a 'Source' column, and sort all entries by timestamp for a chronological activity view.
- Identifying Activity Clusters: Look for periods of intense activity followed by gaps. Dense browsing followed by long gaps may indicate work sessions. Activity outside normal hours may be significant.
- Using Cluster Timelines: Use the Clusters Timeline worksheet to quickly identify bounded research sessions, then drill into History and Downloads for that period.

A.4 Identifying Suspicious Activity

Indicators of potential data exfiltration: downloads of archive files (.zip, .rar), access to cloud storage services, uploads to file sharing sites, and USB/external storage paths in download locations.

Indicators of cover-up attempts: large gaps in History Gaps worksheet, multiple gaps clustered at similar times, clearing data before specific events, or 'Crashed' exit status at unusual times.

Extension-related risks: extensions with broad host permissions, unlisted or developer-mode extensions, extensions whose homepage URLs never appear in History (installed but not researched by user).

A.5 Search Query Analysis

Search Type	Investigative Significance
Technical searches	Skills, interests, projects
Location searches	Places of interest, travel plans
Person searches	Relationships, research on individuals
Legal/medical searches	Personal situations, concerns
Job searches	Employment status, dissatisfaction
Product searches	Purchases, interests

Annex B: URL Pattern Reference for Common Services

Email Services

Service	URL Pattern	Indicates
Gmail Inbox	mail.google.com/mail/u/0/#inbox	Viewing inbox
Gmail Sent	mail.google.com/mail/u/0/#sent	Viewing sent mail
Gmail Compose	mail.google.com/mail/u/0/?compose=	Composing email
Outlook Inbox	outlook.live.com/mail/0/inbox	Viewing inbox
Yahoo Mail	mail.yahoo.com/d/folders/1	Viewing inbox

Cloud Storage

Service	URL Pattern	Indicates
Google Drive	drive.google.com/drive/	Browsing files
Google Drive File	drive.google.com/file/d/	Viewing specific file
OneDrive	onedrive.live.com/?id=	Viewing files
Dropbox	dropbox.com/home	Viewing files
Dropbox Shared	dropbox.com/sh/	Accessing shared folder
Box	app.box.com/	Viewing files
iCloud	icloud.com/iclouddrive	Viewing files

File Sharing / Transfer

Service	URL Pattern	Indicates
WeTransfer	wetransfer.com	File transfer
MediaFire	mediafire.com/file/	File access
Mega	mega.nz/	File access
Google Drive Share	drive.google.com/file/d/[id]/view	Shared file

Annex C: Timestamp Quick Reference

All 'Decoded' timestamp columns are in UTC (Coordinated Universal Time). To convert to local time, note the UTC timestamp, apply the appropriate timezone offset, and account for Daylight Saving Time if applicable.

Worksheet	Primary Timestamp Field	What It Represents
History	Decoded visit_time	When page was visited
Downloads	Decoded start_time	When download began
Downloads	Decoded end_time	When download completed
Cookies	Decoded creation_utc	When cookie was created
Cookies	Decoded last_access_utc	When cookie was last used
Login Data	Decoded date_created	When credential was saved
Login Data	Decoded date_last_used	When credential was last used
Autofill	Decoded date_created	When value was first entered
Autofill	Decoded date_last_used	When value was last used
Bookmarks	Converted Date added	When bookmark was created
FavIcons	Decoded last_updated	When favicon was refreshed
Shortcuts	Decoded last_access_time	When shortcut was used
Clusters Contents	visit_time (decoded UTC)	When cluster visit occurred
Clusters Timeline	first_visit_time (decoded UTC)	First visit in cluster
Clusters Timeline	last_visit_time (decoded UTC)	Last visit in cluster
Clusters Search Term	earliest_visit_time (decoded UTC)	Earliest visit with that search term
WebAssist	decoded last_visited_time	When page was visited (Edge)